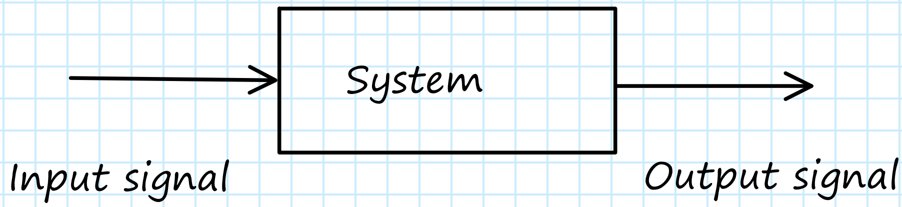
Lctr 2: DT signals and systems



Learning outcomes:

- LO2. Express signals in terms of basic signals
- LO3. Determine periodicities of DT signals
- LO4. Determine linearity, time invariance, causality, and stability of DT systems

LO2. Express signals in terms of basic signals

- Def: DT signal/sequence x

$$x : \text{Integers} \rightarrow \text{Complex}$$

DT signal x is a function that maps integers to complex values

$x[n]$: the value of x at index n.

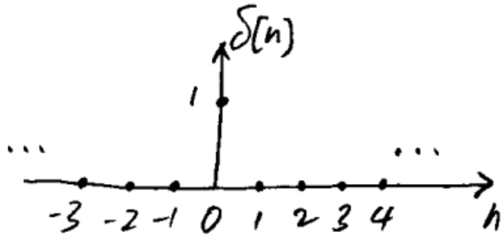
$\{x[n]\}$: the collection of all values of x.

We usually say $x[n]$ for both the entire signal/sequence and the value at index n.

- Basic signals

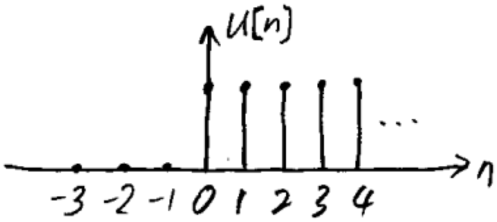
· The DT impulse of unit sample

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases}$$



· The DT unit step

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$



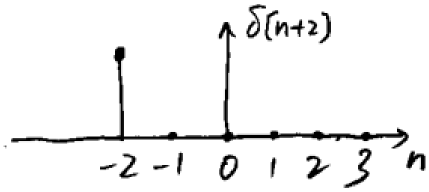
- Basic systems

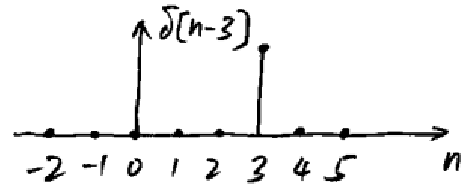
  - Shift

$$x[n] \longrightarrow \boxed{\begin{matrix}\text{shift} \\ n_0\end{matrix}} \longrightarrow x[n-n_0]$$

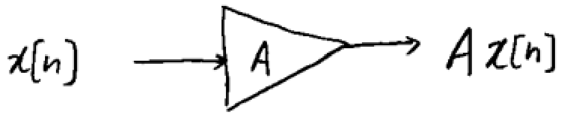
$n_0 > 0$: delay by n_0 ticks (shift right)

$n_0 < 0$: advance by $-n_0$ ticks (shift left)

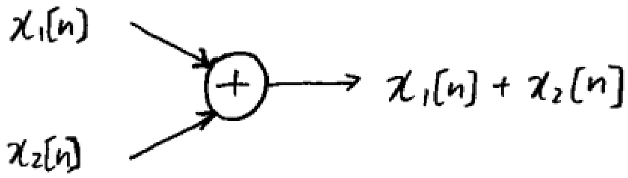




  - Scale



  - Add



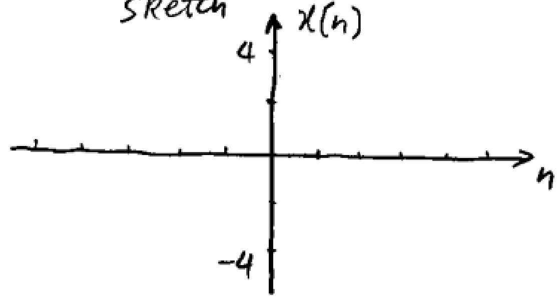
- Other signals

  - Any DT signal can be represented as a <u>sum</u> of <u>scaled</u> and <u>delayed</u> unit impulses

Example:

$$x[n] = \begin{cases} -4, & n=-2 \\ 3, & n=-1 \\ -3, & n=1 \\ 4, & n=2 \\ 0, & \text{otherwise} \end{cases}$$

sketch



Formula:

$x[n] =$

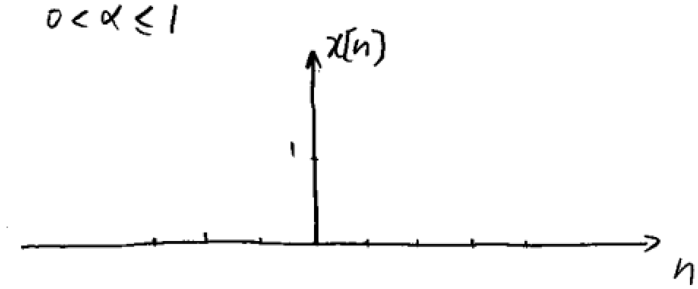
$\square$

- Exponential sequences

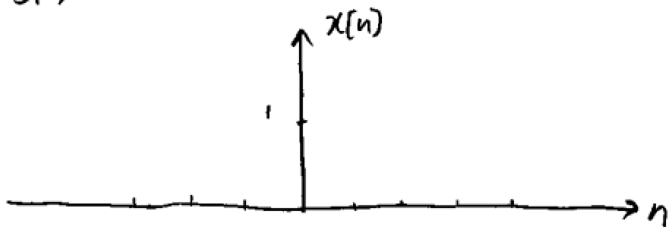
$$x[n] = \alpha^n$$

— When α is real

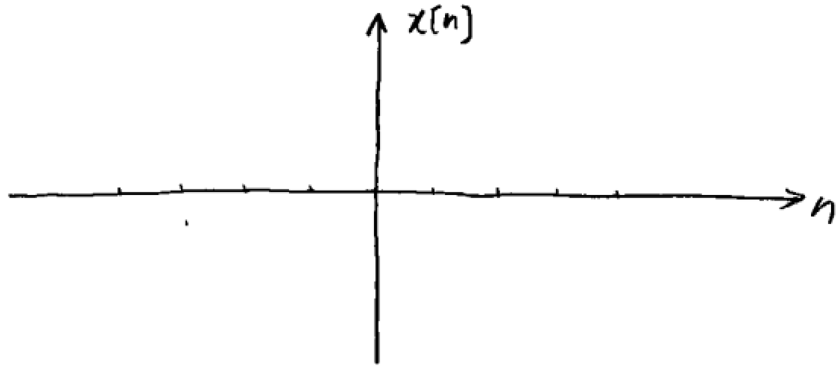
① $0 < \alpha \leq 1$



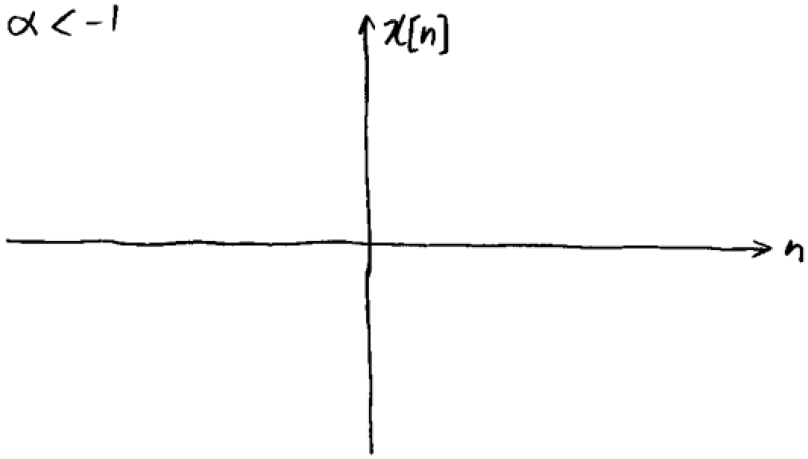
② $\alpha > 1$



③ $-1 \leq \alpha \leq 0$



④ $\alpha < -1$



– When α is complex

$\alpha = |\alpha| e^{j w_0}$

$\rightarrow x[n] = \alpha^n = |\alpha|^n e^{j w_0 n}$

$= |\alpha|^n (\qquad + j \qquad).$

LO3. Determine periodicities of DT signals

- Def: A DT signal x[n] is said to be periodic with period N when

$$x[n+N] = x[n], \quad \forall n \in \mathbb{I}, \; N \in \mathbb{I}$$

- Question:

For what values of w_0 can $e^{jw_0 n}$ be periodic with period $N \in \mathbb{I}$?

Soln: $e^{jw_0(n+N)} = e^{jw_0 n}$

$\Rightarrow \quad e^{jw_0 N} =$

$\Rightarrow$

$\Rightarrow$

<u>Result:</u> A DT Sinusoid or complex exponential is peoridic with period N when $\exists\, k \in \mathbb{I}$ such that

$$N = \frac{2\pi}{w_0} k \in \mathbb{I}$$

- Def: The smallest value of the above N is the fundamental period.

Example: Is $e^{j\frac{3\pi}{4}n}$ periodic? If yes, what is the fundamental period?

Soln:

(a) $e^{j\frac{3\pi}{4}(n+N)} = e^{j\frac{3\pi}{4}n}$

requires

which requires

Which can be satisfied if

$\Rightarrow e^{j\frac{3\pi}{4}n}$

(b) $\frac{3\pi}{4}N = 2\pi k$

$\Rightarrow N =$

For $k=$, $N=$, which is the fundamental period.

$\square$

Question: Is $e^{j\frac{\pi}{5}n}$ periodic? Period?

Is $\cos(n)$ periodic? period?

LO4. Determine linearity, time-invariance, causality, and stability of DT systems

$$x[n] \longrightarrow \boxed{T\{\ \}} \longrightarrow y[n] = T\{x[n]\}$$

$T\{\ \}$ is the <u>rule</u> by which $y[n]$ is formed using $x[n]$

- *Linearity: T{ } is linear iff*

$$T\{a\,x_1[n] + b\,x_2[n]\} = a\,T\{x_1[n]\} + b\,T\{x_2[n]\}$$

- Remarks: To prove/disprove linearity, use the rule to see what the output is for each of

$$x_1[n], \quad x_2[n], \quad \text{and } a\,x_1[n] + b\,x_2[n]$$

as input; then see if the above linearity condition is observed.

- Example: Is $y[n] = T\{x[n]\} = \dfrac{1}{x[n+1]}$ linear?

Soln:

$$x[n] = x_1[n] \Rightarrow y[n] = y_1[n] = \frac{1}{x_1[n+1]}$$

$$x[n] = x_2[n] \Rightarrow y[n] = y_2[n] = \frac{1}{x_2[n+1]}$$

$$x[n] = x_3[n] = a\,x_1[n] + b\,x_2[n]$$

$$\Rightarrow y[n] = y_3[n] = \frac{1}{a\,x_1[n+1] + b\,x_2[n+1]}$$

$$y_3[n] \neq a\,y_1[n] + b\,y_2[n]$$

$\Rightarrow T\{\ \}$ is not linear.

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- Time invariance: T{ } is time invariant iff

$$T\{x[n-n_0]\} = T\{x[n]\} \text{ shifted by } n_0.$$

- Remarks: The same basic approach as above, but only need one test: the output of the delay input vs the delayed output of the input.

- Example: Is $y[n] = T\{x[n]\} = \frac{1}{x[n+1]}$ time invariant?

Soln: $x[n] = x_1[n] \Rightarrow y[n] = y_1[n] = \frac{1}{x_1[n+1]}$

$$= y_1[n-n_0] = \frac{1}{x_1[n-n_0+1]}$$

$x[n] = x_1[n-n_0] = x_2[n] \Rightarrow$

$$y[n] = y_2[n] = \frac{1}{x_1[n-n_0+1]}$$

$\Rightarrow$ Time invariant.

□

- Causality: T{ } is causal iff y[n] = T{x[n]} is independent of future values of x[n]; i.e., y[n] only depends on

$$x[n],\ x[n-1],\ x[n-2], \ldots$$

But not

$$x[n+1],\ x[n+2], \ldots$$

- Example: Is $y[n] = T\{x[n]\} = \frac{1}{x[n+1]}$ causal?

Soln:

□

- BIBO (Bounded-input bounded-output) stability:
  T{ } is BIBO stable iff

$$|x[n]| \le M_x < \infty \quad \Rightarrow \quad |y[n]| \le M_y < \infty, \quad \forall n \in \mathbb{Z}.$$

· Example: Is $y[n] = T\{x[n]\} = \frac{1}{x[n+1]}$ BIBO stable?

Soln:

Example: Is $y[n] = T\{x[n]\} = \cos(\omega_0 n) \cdot x[n]$

linear, time invariant, causal, and BIBO stable?

Soln: